

Program 4:

Integrate Git with Docker for source-controlled application build

07/11/2025

Project Directory:

- |-Dockerfile
- |-package.json
- |-package-lock.json
- |-index.js

Program code and steps for execution:

Step 1: Create a folder with the name prog4 for the program 4 execution

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~$ cd mylab/  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab$ mkdir prog4  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab$ cd prog4/
```

Step 2: Now move to the created folder and do the npm initialization → npm init -y

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ npm init -y  
Wrote to /home/chinmayi-m-h/mylab/prog4/package.json:
```

```
{  
  "name": "prog4",  
  "version": "1.0.0",  
  "description": "",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "type": "commonjs"  
}
```

Step 3: Now create an index.js file

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ sudo nano index.js  
[sudo] password for 1RV24MC030_chinmayi_m_h:  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ cat index.js  
const express=require("express");  
const app=express();  
const PORT=3000;  
  
app.get("/", (req,res)=>{  
  res.send("Hello this is the GIT and DOCKER integration");  
});  
  
app.listen(PORT, ()=>{  
  console.log(`Server is running at http://localhost:${PORT}`);  
});
```

Step 4: Now install the express → npm install express

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ npm install express
added 68 packages, and audited 69 packages in 3s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

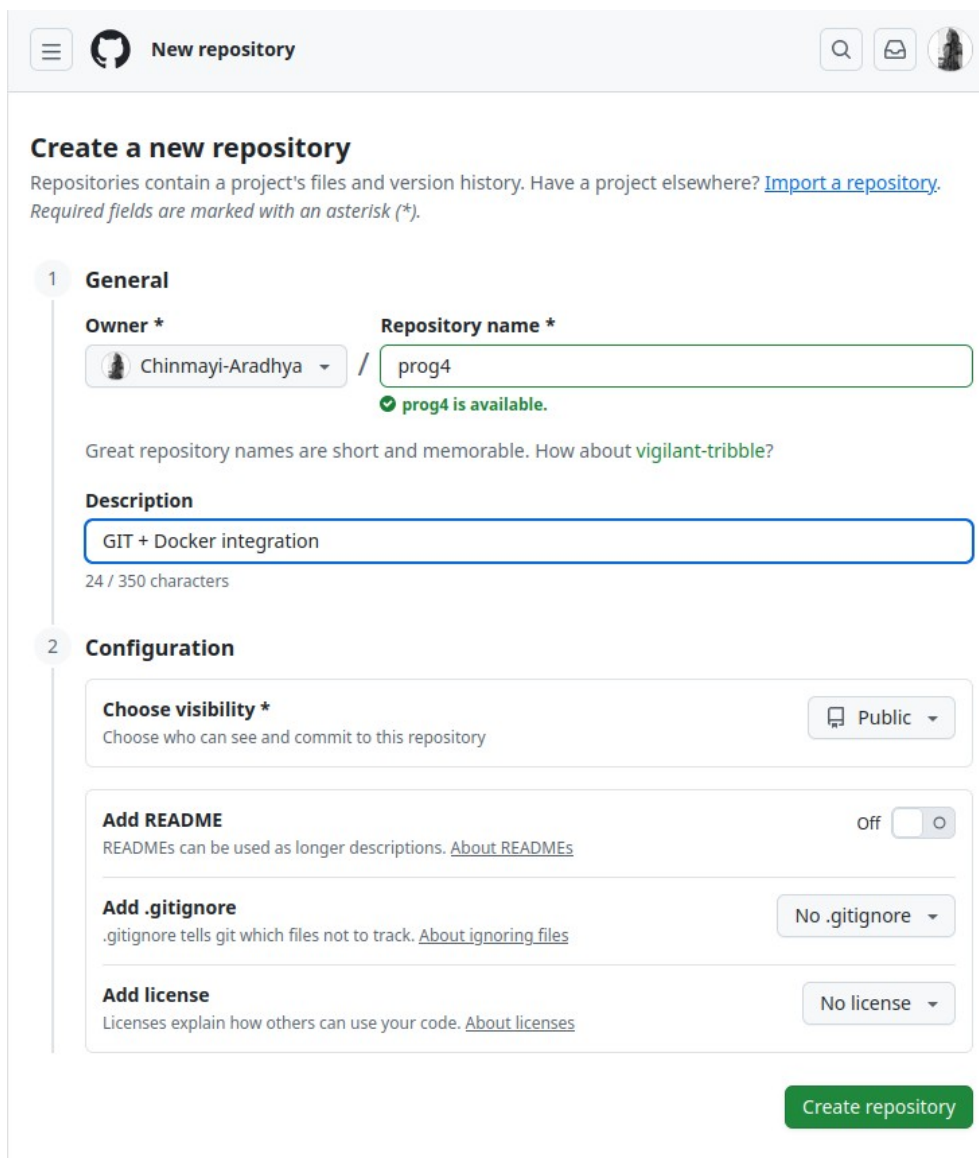
Step 5: The Dockerfile is

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ sudo nano Dockerfile
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ cat Dockerfile
# Create a Dockerfile with 2 stages

# Stage 1: Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# Stage 2: Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
```

Step 6: Open the browser and login to your github account and create a new repository



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)
Required fields are marked with an asterisk (*).

1 General

Owner * Chinmayi-Aradhya / **Repository name *** prog4
✓ prog4 is available.

Great repository names are short and memorable. How about [vigilant-tribble?](#)

Description
GIT + Docker integration
24 / 350 characters

2 Configuration

Choose visibility *
Choose who can see and commit to this repository
Public

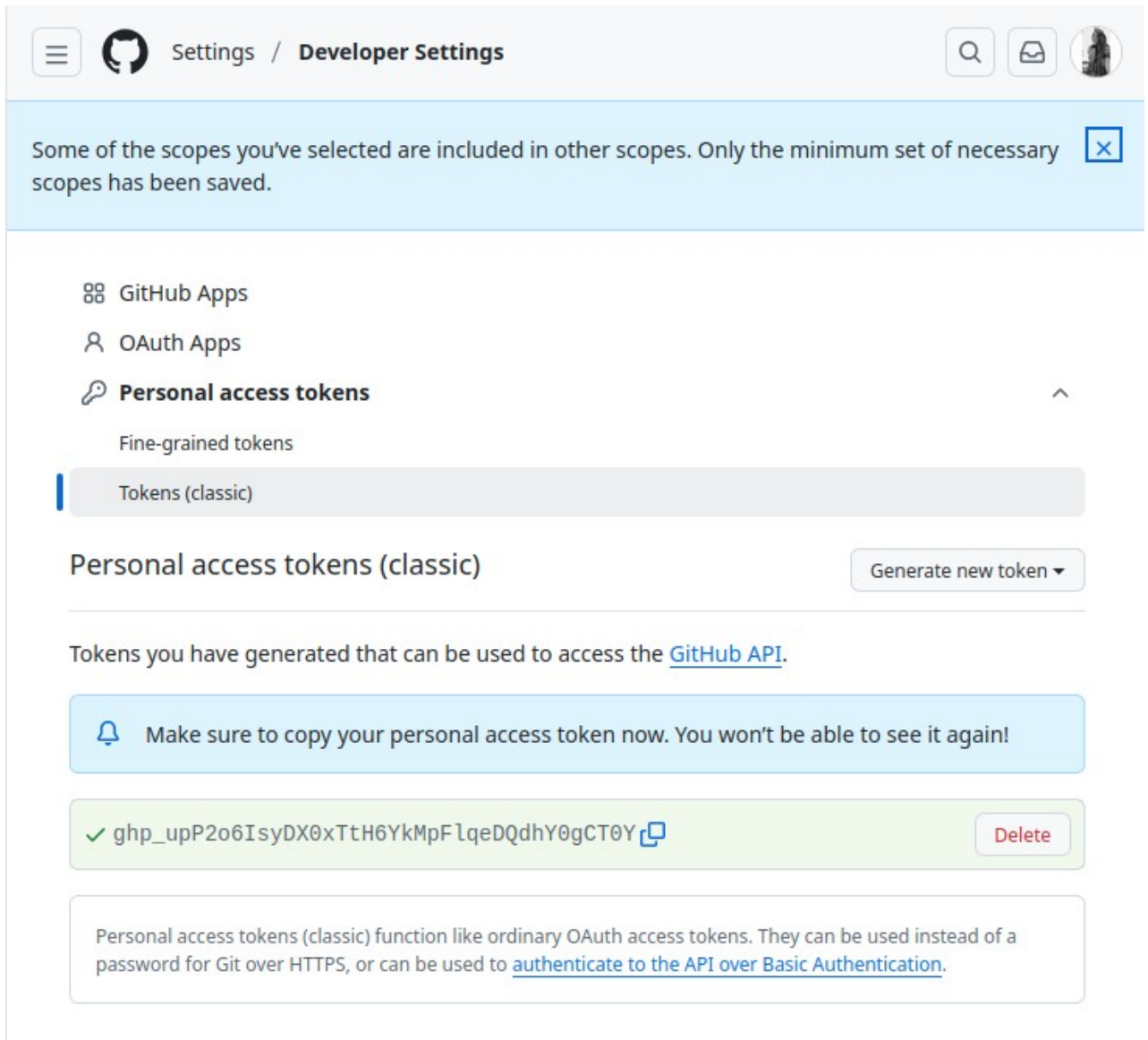
Add README
READMEs can be used as longer descriptions. [About READMEs](#)
Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#)
No license

Create repository

Step 7: In the github home page of your account click on the profile and go to Developer Settings and create a Personal Access Token, copy and keep it somewhere as it is visible only once



Step 8: Again back to terminal and initialize the git locally → git init

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/chinmayi-m-h/mylab/prog4/.git/
```

Step 9: Add all the files which are present in that folder to the git using the command → git add .

Step 10: Perform the initial commit to the main branch

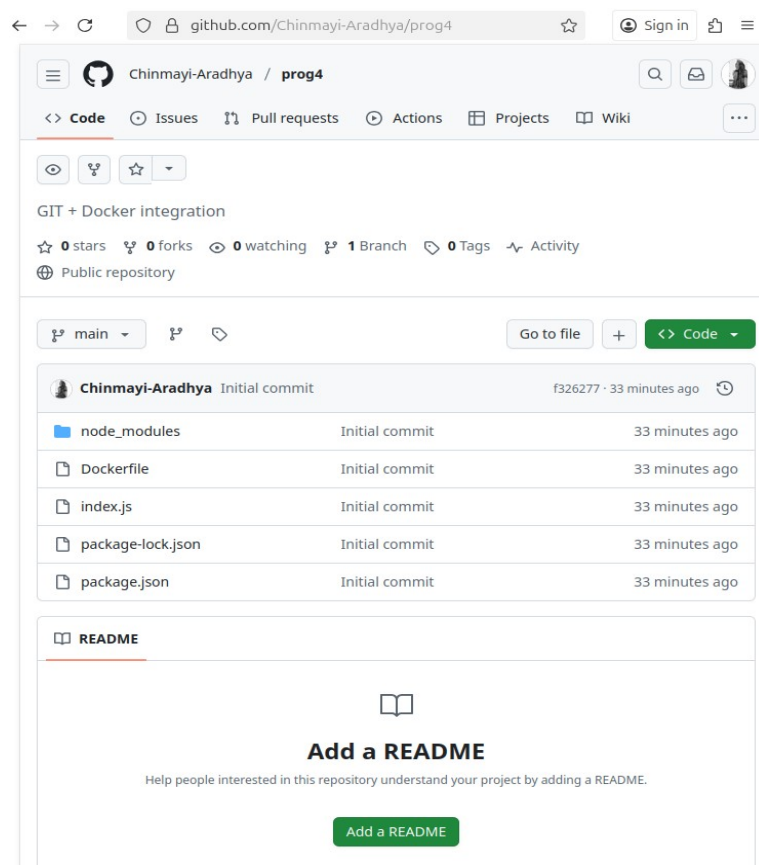
```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ git add .
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ git commit -m "Initial commit"
[master (root-commit) f326277] Initial commit
636 files changed, 73047 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
```

Step 11: Once you created the repository the repo will provide the ssh link to remotely add the git repo to through terminal which looks like → git remote add origin <http://github.com/><user-name>/<repo-name>.git>

Step 12: Now select the branch to which all the files of the folder must need to be moved then push the files to the branch

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ git remote add origin https://github.com/Chinmayi-Aradhya/Prog4.git
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog4$ git branch -M main
git push -u origin main
Username for 'https://github.com': Chinmayi-Aradhya
Password for 'https://Chinmayi-Aradhya@github.com':
Enumerating objects: 722, done.
Counting objects: 100% (722/722), done.
Delta compression using up to 16 threads
Compressing objects: 100% (667/667), done.
Writing objects: 100% (722/722), 734.56 KiB | 7.73 MiB/s, done.
Total 722 (delta 140), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (140/140), done.
remote: This repository moved. Please use the new location:
remote: https://github.com/Chinmayi-Aradhya/prog4.git
To https://github.com/Chinmayi-Aradhya/Prog4.git
* [new branch] main -> main
```

Step 13: Now the repo looks like



The screenshot shows the GitHub web interface for the repository 'Chinmayi-Aradhya / prog4'. The repository is public and has 0 stars, 0 forks, and 0 watchers. It has 1 branch (main) and 0 tags. The file list shows the following files, all from the initial commit (f326277) 33 minutes ago:

File	Commit	Time
node_modules	Initial commit	33 minutes ago
Dockerfile	Initial commit	33 minutes ago
index.js	Initial commit	33 minutes ago
package-lock.json	Initial commit	33 minutes ago
package.json	Initial commit	33 minutes ago

Below the file list, there is a section for the README, which currently says 'Add a README' and 'Help people interested in this repository understand your project by adding a README.' with a green 'Add a README' button.